

PRODUCTION BASELINE v1.0

MASS-APP-015-V02 - Plugin Lifecycle, Installation & Dependency Management

MASS-APP-015-V02

EWO-MASS-APP-015-V02

LCO-009-B regenerated canonical package: all eight lifecycle roles are distinct, including Security Approver and Entitlement Approver. Substantive scope is unchanged.

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1. Purpose

V02 converts a V01-validated plugin version into a controlled tenant installation. It owns lifecycle requests, eligibility snapshots, dependency resolution, immutable installation plans, independent approvals, tenant installation records, configuration/data migration coordination, health findings, rollback evidence, and safe-removal records. Installation never grants capability invocation authority.

2. Permanent Architecture and V1 Implementation

Permanent architecture separates declared package truth, evaluated target truth, human decisions, execution coordination, and immutable evidence. V01 remains authoritative for manifests, versions, capabilities, and declared dependencies. V02 records the exact V01 version and digest used by each plan.

V1 implements organization, workspace, project, and user scopes; required, optional, peer, and conflicting dependencies; deterministic version selection; explicit approval; idempotent lifecycle commands; configuration migration; application-owned data migration references; readiness checks; rollback; repair; removal; orphan analysis; deprecation; and end-of-support controls.

V1 does not execute capabilities, duplicate deployment-engine ownership, build a storefront, silently install dependencies, or remove application-owned data.

3. Role Mapping

Role	Baseline Mapping	Authority
Installation Viewer	Viewer	Read authorized plans, state, health, and evidence
Installation Requester	Contributor	Request installation or lifecycle change
Plugin Operator	Contributor extension	Execute an approved, bound plan through platform gateways
Lifecycle Steward	Steward	Resolve dependencies, assess compatibility, and recommend recovery
Installation Approver	Administrator	Approve or reject an immutable plan
Security Approver	Administrator specialization	Approve plans with security-sensitive grants
Entitlement Approver	Administrator specialization	Confirm license and entitlement references
Executive Approver	Executive authority	Decide exceptional risk explicitly requiring executive authority

The requester, plugin, publisher, and execution principal cannot approve their own plan. Approval binds the manifest digest, resolved graph, target scope, permissions, migration steps, and rollback plan.

4. Platform Consumption Map

Source	Consumption	Boundary
APP-015 V01	Published manifest, version digest, capability/dependency declarations, eligibility	V01 remains registry authority
ENG-003 / ENG-004	Principal identity, role, authorization, separation of duty	Platform-owned
ENG-005 / ENG-006	Lifecycle events and approved workflow coordination	V02 does not own generic orchestration
ENG-011	Audit, health observations, correlation identifiers	Evidence is never silently dropped
ENG-012	Tenant configuration references and governed secrets	No secret value is copied into V02 evidence
ENG-015	Versioned API and event contracts	Contract authority remains platform-owned
ENG-016	Environment and deployment execution contracts	V02 plans and requests; deployment engine executes
ENG-018	License and entitlement references when available	Absence blocks entitlement-required activation
ENG-027	Provenance and information lineage	Digests and source references remain traceable
APP-021 / APP-022	Administration and security governance when available	No ownership assumed before published contracts

5. Gateway Inventory

Gateway	Purpose	Failure Behavior
PluginRegistryGateway	Resolve V01 manifest, version, digest, declarations	Reject missing, unpublished, invalid, or changed version
IdentityGateway	Resolve actor and approver identity	Fail closed
AuthorizationGateway	Check lifecycle action and scope grant	Fail closed without widening permissions
WorkflowGateway	Coordinate approval and execution steps	Pause plan; preserve completed-step evidence
EventGateway	Publish lifecycle events through an outbox	Retry; never erase unpublished event
AuditGateway	Correlate immutable audit evidence	Lifecycle decision fails if durable audit cannot be recorded
ConfigurationGateway	Resolve approved configuration and secret references	Block readiness; never materialize secret values in evidence
DeploymentGateway	Execute environment/deployment operations	Record failure and enter recovery; V02 does not impersonate executor
EntitlementGateway	Verify license/entitlement reference	Block activation when required evidence is absent or expired
LineageGateway	Bind package, plan, execution, migration, and rollback provenance	Mark evidence incomplete and block irreversible transition
AdministrationGateway	Resolve organization/scope policy when available	Use explicit platform default only when contract permits
SecurityGovernanceGateway	Obtain security decision when available	Security-required plan remains awaiting approval

6. Installation Request and Planning Lifecycle

1. A requester submits plugin version, target scope, requested permissions, configuration references, and an idempotency key. 2. V02 snapshots the V01 manifest digest and verifies publication, tenant eligibility, platform/application/environment prerequisites, entitlement, and declared permissions. 3. The resolver builds a deterministic graph from declared dependencies only. It selects versions, explains alternatives, identifies optional omissions, verifies peer satisfaction, and records conflicts. 4. V02 produces an immutable installation plan containing exact versions/digests, topological steps, approvals, migrations, health gates, failure compensations, and rollback targets. 5. Independent approvers decide the bound plan. Any material change invalidates approval and creates a new revision. 6. An authorized operator executes each idempotent step through its owning gateway. Step evidence is append-only. 7. Activation occurs only after required steps, grants, entitlement, migration checkpoints, and health/readiness gates pass.

7. Dependency Resolution Rules

- Required: must resolve to exactly one compatible published version in scope or the plan is blocked.
- Optional: may be omitted; omission and capability impact are recorded. An optional dependency is never installed silently.
- Peer: must already exist or be explicitly included at a compatible version and compatible scope; V02 does not own or silently replace it.
- Conflict: any installed or planned version matching the conflict range blocks approval until a human chooses a documented alternative.
- All ranges use the version grammar declared by V01. Exact selection is deterministic: retain a healthy compatible installed version; otherwise choose the highest published eligible version not deprecated beyond policy.
- Cycles among required dependencies are invalid. Optional-only cycles are omitted with a finding. Mixed cycles are blocked.
- A dependency scope must be equal to or broader than the dependent scope and permitted by tenant policy.
- Transitive dependencies remain visible in the plan with provenance and reason chains.
- Resolver output is advisory until an independent human approves the complete immutable plan.
- Unavailable APP-021, APP-022, or ENG-018 contracts are not fabricated; prerequisites depending on them remain unresolved.

The canonical rules are also supplied in `MASS-APP-015-V02_Dependency_Resolution_Rules.md`.

8. Plugin Lifecycle State Machine

Primary states are requested, planning, awaiting_approval, approved, installing, installed, activating, active, suspending, suspended, upgrading, rolling_back, repairing, removing, removed, recovery_required, and failed.

Transitions require expected current state, installation revision, approved plan digest, actor authorization, and idempotency key. Repeated commands return the original outcome. Stale revisions fail without side effects. Failure never reports the desired state; it records the completed step boundary and enters `recovery_required` or `failed`.

The authoritative diagram and transition guards are in `MASS-APP-015-V02_Plugin_Lifecycle_State_Machine.mmd`.

9. Installation, Activation, and Suspension

Installation creates tenant-scoped records, verifies package digest, records grants no broader than the manifest and approved tenant grant, applies V02-owned configuration schema changes, requests application-owned data migrations through the owning gateway, and performs readiness checks. `installed` means present but not executable.

Activation requires approved status, satisfied dependencies, completed migrations, healthy required checks, valid entitlement, current security decision, and an explicit authorized action. Runtime invocation remains V03. Suspension prevents new capability invocation through the future runtime contract while retaining installation state, evidence, configuration references, and owned data.

10. Upgrade, Downgrade, Rollback, and Repair

Every upgrade or downgrade creates a new plan comparing source and target digests, permissions, dependencies, configuration schema, application data migration requirements, compatibility, health criteria, and rollback feasibility. Permission expansion requires new approval. A downgrade is rejected when no safe migration/rollback path exists.

Rollback uses a previously approved rollback target and immutable checkpoint. It never invents reverse migrations. Repair revalidates the installed digest, dependencies, configuration references, grants, and health, then replays only explicitly idempotent steps. All actions preserve the prior version and evidence records.

11. Migration Ownership

V02 owns its installation metadata schema and plugin configuration-schema coordination. Each source application or engine owns its data. V02 records a migration contract reference, owner, checksum, preconditions, checkpoint, result, and compensation reference, but does not author or delete another owner's data. Irreversible migrations require explicit warning and approval; absent recovery evidence blocks activation.

Reference DDL: `MASS-APP-015-V02_Migration_Reference.sql`.

12. Health, Partial Failure, and Recovery

Health checks are declared, time-bounded, severity-classified, and attached to a lifecycle stage. Required readiness failures block activation. Degraded non-required checks may proceed only when the approved plan permits it.

Each execution step is pending, running, succeeded, failed, compensated, or skipped_by_approved_plan. On failure, V02 atomically records the failure and durable lifecycle/outbox evidence before recovery. Completed irreversible steps are never hidden. Recovery resumes from the last verified checkpoint or executes the approved compensation sequence. If audit/outbox persistence fails, the state transition fails and remains retryable.

13. Removal and Orphan Handling

Removal requires reverse-dependency analysis, suspension, final evidence capture, V02 metadata/configuration cleanup, and a retained tombstone. V02 never removes application- or engine-owned data. Required dependents block removal. Optional dependents receive explicit impact findings. Dependencies installed solely for the removed plugin become orphan candidates, never automatic deletions. Humans approve each candidate removal plan.

End-of-support and deprecation findings are visible in planning and health. An active unsupported version generates an advisory and remediation options; it is not silently upgraded or disabled unless separately authorized by policy and human action.

14. Organization and Scoped Installations

Organization installations may satisfy narrower workspace, project, or user dependencies only when the manifest, tenant policy, grants, and configuration isolation permit inheritance. Scoped installation records always retain their exact scope_type and scope_id. Promotion or widening is a new plan and approval, not an update in place.

15. API and Event Contracts

The API inventory is MASS-APP-015-V02_API_Inventory.csv. All mutation endpoints require Idempotency-Key, expected installation revision where applicable, actor authorization, and correlation identifiers. Contract examples are in MASS-APP-015-V02_Installation_and_Rollback_Contract_Examples.json.

Events include plugin.installation.requested, plugin.installation.plan-created, plugin.installation.approved, plugin.installed, plugin.activation.requested, plugin.activated, plugin.suspended, plugin.upgrade.started, plugin.rollback.completed, plugin.recovery.required, and plugin.removed. Events are written to the tenant outbox in the same transaction as the authoritative state change.

16. Security and Database Integrity

All 17 tenant-owned tables use DEFAULT gen_random_uuid() UUID primary keys, UNIQUE(id, tenant_id), composite tenant-safe foreign keys, and auth.jwt()-derived RLS. Plan approvals, lifecycle decisions, append-only version snapshots, step evidence, migration checkpoints, rollback evidence, health observations, and audit/outbox records become immutable at their governing state. Triggers prohibit self-approval, stale-state mutation, direct state rewriting, and destructive evidence updates.

Secret values, license material, and deployment credentials are referenced but never copied into V02 records. Permissions equal the intersection of declared permissions, approved plan permissions, tenant policy, and actor grant.

17. Constitutional Boundary Statement

V02 may validate, plan, compare, recommend, coordinate an authorized lifecycle action, and preserve evidence. It cannot expand privilege, install undeclared dependencies, self-approve, imply runtime execution authority, duplicate deployment ownership, bypass license/security/privacy controls, remove another owner's data, suppress failures, or make an irreversible change without the approved bound plan.